

IS1128 IT, Management, Organization

Module focuses on information technologies (IT) in organizations and the interplay between IT, work, management, and organizations.

- Multifaceted roles of IT, support communication, collaboration, and organizational improvements in operations, planning, and decision making.
- Opportunities for IT-enabled innovations and issues involving information systems (IS) adoption and deployment.
- Impacts of modern IT and related artificial intelligence (AI) technologies on knowledge workers, teamwork, work design, management practices, and organization.

1. Impact of IT/IS on Businesses

Global spending on information technology (IT) and IT services steadily increasing. Organizational, management, and cultural changes required for firms to derive full business value from IT investment. But does not guarantee good returns. Overall impacts on organization (structure, culture, team capability), technology, management.

Strategic business objectives

- **Survival:** Information technologies as necessity of business. *Innovation Diffusion Model:* Roger's bell curve adoption of technology (Innovators, early adopters, early/late majority, laggards).
- **Operational excellence:** Improvement of efficiency and effectiveness to attain higher profitability. (e.g. accounting systems, marketing forecasters, production management software). *Automating (faster), Learning (better - exploitation and exploration), Strategizing (smarter than rivals)*. E.g. Walmart VMI Retail Link reduces bullwhip effect.
- **New products/service/business model:** E.g. sharing economy, Netflix.
- **Customer and supplier intimacy:** Customers served well become repeat customers. Close relationships with suppliers result in lower costs. E.g. investment in CRM solutions.
- **Improved Decision Making:** Real-time, accurate information. Better forecasts, planning, response time. E.g. Jooz real time metrics, Starbucks GIS.
- **Competitive advantage:** Advantage over competitors. Better performance, price competitive. Sometimes, create brand image of innovation at initial loss.

Overall, growing interdependence between information technology and implementing corporate strategies to achieve corporate goal. We assume a **sociotechnical view**, where optimal organizational performance achieved by jointly optimizing both social and technical systems used in production. Helps avoid purely technological approach.

2. IS and Organizations

Features of organizations relevant to build, use IT/IS successfully: Org. structure, Business model, Business process, Business/IT alignment.

Organization

- **Technical:** A stable, formal social structure that takes resources from the environment and processes them to produce inputs. Formal legal entities with internal rules, procedures.
- **Behavioral:** A collection of rights, privileges, obligations, and responsibilities delicately balanced through conflict and conflict resolution.
- Organizations are Bureaucracies with clear-cut divisions of labor and specialization, adhere to *principle of efficiency* to maximizing output using limited inputs.

Organizational Structure

The formal configuration between individuals and groups regarding the allocation of tasks, responsibilities, and authority within the organization. We define 5 basic kinds:

- **Entrepreneurial:** Simple structure, Fast-changing environment, Major decision made by one or two key personnel, Direct supervision, e.g., small start-up business.
- **Machine Bureaucracy:** Centralized management/decision making, Stable environment, High degree of formalization and work specialization, Standardization of work processes, Many levels in the chain of command, Goal: achieve internal efficiency, e.g., midsize manufacturing firm, government agencies.
- **Divisionalized Bureaucracy:** Multiple machine bureaucracies, Each producing a different product or service, Decentralized decision-making at the divisional level, Little coordination among separate divisions, Topped by a central headquarters, e.g., Fortune 500 firms with multiple product lines.
- **Professional Bureaucracy:** Knowledge-based organization, Standardization of skills, Goods and services depend on professionals' expertise, Highly trained professionals provide nonroutine services, Formalized but decentralized to give autonomy, Department heads with weak centralized authority, Small top management, Few middle managers, Goal: Innovate and provide high-quality services, e.g., law firms, universities, hospitals.
- **Adhocracy:** Low formalization and decentralization, Task-force organization responding to a rapidly changing environment, Engages in nonroutine tasks and uses sophisticated technology, Goal: Innovation and rapid adaptation, Large groups of specialists in short-lived multidisciplinary teams, Weak central management, e.g., NASA, Pixar Animation Studios.

Information systems often reflect organizational structure, such as centralization or decentralization.

Business Model

A business model specifies how a company will create, deliver, and capture value. What does company do? How unique? Revenue, Cost, key resources / activities / partners needed? Business Model Canvas.

- **Business Processes:** Logically related set of activities/steps that define how specific business tasks are performed. Flows of material, information, knowledge, tied to functional area or be cross-functional
- Businesses can be seen as collection of business processes, business processes may be assets or liabilities.
- Focus on alignment of IT strategy with business's strategy. Continuous process of adjusting business goals and IS architecture to achieve business objectives, support/enable business transformation.

3. IS and Competition Strategies

Focus on how Porter's competitive forces model, the value chain model, and SWOT analysis help companies develop competitive advantages (CA) using IT/IS. Competition strategies for organizations to handle disruptive technology and disruptions.

A. Where to Compete: Analyzing Competitive Forces 5 Forces

Analyze competitive forces to focus resources effectively. Michael Porter's Five Forces Model provides a general view of a firm, competitors, and environment.

- **Rivalry among existing competitors:** High intensity if competitors are numerous or roughly equal in size and power. Slow Industry growth, High exit barriers. Rivals are highly committed with leadership aspirations. Firms struggle to read each other's signals due to unfamiliarity.
- **Threat of new entrants:** Barriers to entry include Supply-side economies of scale, Demand-side benefits of scale. Customer switching costs. Capital requirements. Incumbency advantages independent of size. Unequal access to distribution channels. Restrictive government policies.
- **Bargaining power of suppliers** Suppliers are powerful if: More concentrated than the industry they sell to. Do not depend heavily on the industry for revenue. Switching costs in changing suppliers are high. Products differentiated. No substitutes. Can credibly threaten forward integration.
- **Bargaining power of buyers** Buyers have strong negotiation power if: few buyers or large purchase volume. Products standardized or undifferentiated. Switching costs low. Can credibly threaten backward integration.
- **Threat of substitute products or services** High threat if substitutes offer attractive price-performance trade-off and switching costs low.

Type of Business

- **Pipeline Business:** Creates value by controlling a linear series of activities. Follows the classic value chain model. Inputs (e.g., materials from suppliers) undergo transformations into a finished product with higher value. - E.g. Nike, Dell.
- **Platform Business** Facilitates interactions between producers and consumers. *Critical asset:* Community. Resources of participating groups. Strategic focus shifts from: Resource control to orchestrating resources, Internal process optimization to facilitating external interactions, End-user value to maximizing ecosystem value. - E.g. Gojek, Airbnb.

Network Effects

- **Direct Network Effects (One-Side Effects):** Utility user receives increases as the number of users grows. E.g. Communication platforms. *Differentiation:* Capacity-constrained vs. capacity-unconstrained assets.
- **Indirect Network Effects (Cross-Side Effects):** Value of a service for one user group rises when another user group expands. Critical for multi-sided platforms.

Market Concentration

- **Concentration Ratio (CR):** Measures the combined market share percentage of the largest firms in an industry. - Typically calculated for the top three or four firms (e.g., CR4). Market types: Monopoly: CR1 100%, Oligopoly: CR5 > 60%, Pure competition: CR4 0%.
- **Herfindahl-Hirschman Index (HHI):** Derived by summing the squares of all firms' market shares. Higher values equal dominance by large players. Value ranges: Decimal format: (0,1]. Whole number format: (0,10000].
- Sample reference: HHI < 1500 non-concentrated, HHI 1500-2500 moderate, HHI > 2500 is highly concentrated.

B. How to Compete: Choosing Generic Strategy

- **Low cost leadership strategy:** Lower price than competitors, greater efficiency. Keep tight control of overheads and maximizing the cost benefit of industrial experience and new technology. Avoid unprofitable/marginal customer accounts, minimize running cost or investment in processes seen as ancillary, such as R&D, advertisement, and customer service.
- **Product differentiation** New products/services, Innovation, high start-up and running cost.

- **Market Niche** Specific market focus.

C. Identify Resources and Capabilities

- **Resources:** Organization's specific assets utilizable to achieve cost or product differentiation E.g., proprietary technology, brand equity, a loyal and established customer base
- **Capability:** The organization's ability to leverage these resources in the marketplace.
- **Distinctive Competencies:** innovation, agility, quality, or low cost. The resources and capabilities develop these competencies, that contribute to some sustained competitive advantages.

SWOT Analysis

Strengths, Weaknesses, Opportunities, and Threats. SWOT analysis is a framework to help assess and understand the internal and external forces that may create opportunities or risks for an organization.

- **Strengths:** Traits giving advantage over competitors
- **Weaknesses:** Traits specific to the business that put it at disadvantage.
- **Opportunities:** Elements in external environment that allow formulation, implementation of strategies.
- **Threats:** Threats in external environment that could endanger business, CA, profits. (Not specific)

D. Identify How to Compete: Analyzing the Value Chain

- Organizations as series of activities adding value to products or services (value chain). Analysis helps determine where value is added and the costs incurred. Highlights areas where competitive strategies can be applied.
- Differentiates between:
 - **Primary activities** (e.g., inbound logistics, operations, marketing, sales, service).
 - **Support activities** (e.g., firm infrastructure, HR, technology development, procurement).
- IS can improve efficiency, strengthen customer/supplier relationships. E.g. through benchmarking, industry best practices.

Disruptive Technologies

Technologies that introduce sweeping changes to industries and markets by outperforming existing products. Provides substitute products that perform as well or better, and drives transformation across industries. E.g. Internet, IoT, Mobile payments, Generative AI (GenAI). Need to adapt to digital disruptions.

Adaptation Strategies

- **Fight Back:** Compete directly through new units, acquisitions, or joint ventures. E.g. Carmakers shifting towards electric vehicles.
- **Double Down:** Strengthen existing advantages and focus on core business. (Think it is a fad, ignore, low ROI)
- **Retrench:** Yield ground to new entrants and conduct survival tactics. E.g. Mergers, acquisitions, lobbying.
- **Move Away:** Shift towards new business opportunities.

Two Strategic Dimensions for these adaptation strategies: Existing Market vs. New Market and Offensive Strategy vs. Defensive Strategy. Companies always end up doing some hybrid.

- **Successful Adaptations:** Business innovation, quick to adapt.

- **Failed Adaptations:** Failure to embrace new business models enabled by digital disruption. Business myopia focus too much on existing revenue streams.
- **Considerations:** What is the value proposition? What new opportunities does disruption open up (business/tech trend)? What capabilities needed to realize opportunities?

4. IS Strategic Deployment

Factors organizations need to consider in IT planning and implementation. Includes system acquisition strategies, system development/delivery methods, system conversion strategies.

- **Internally:** Consider perspective of Business/IT Alignment, Organizational, managerial IT features.
- **Externally:** PESTLE analysis. (Political, Economic, Sociological, Technological, Legal, Environment). E.g. govt regulations, inflation rate, religion, laws, env. concerns.

Feasibility Analysis

- **Organizational Feasibility:** how well solution fits within the organization's strategic planning objectives and goals.
- **Political Feasibility:** how well solution fits within the organization's political environment.
- **Technological Feasibility:** how well solution supportable given organization's current technology infrastructure and resources, (hardware, software, networking, personnel).
- **Behavioral/Operational Feasibility:** how well fits within the organization's culture and the extent to which users are expected to accept the solution. (E.g. Amazon surveillance of workers)
- **Economic Feasibility:** net economic benefits of a particular solution: Cost-Benefit Analysis (CBA)

System Acquisition Strategies

- **Internal Acquisition:** In-house development. Optimal if we need full control of the system. *Challenges include* limited IT staff, IT staff having a limited skill set, overworked IT staff, and problems with performance of IT staff.
- **External Acquisition:** External acquisition of a prepackaged system. Commercial off-the-shelf software package (E.g. Cloud solutions). Outsourcing system development for Customization.
- **Acquisition Strategies:** Important to carefully consider the acquisition strategy. *Strategic tasks* enhance the organization's competitiveness, while critical tasks support core and essential business operations. Routine tasks should also be evaluated.
- Key factors in acquisition decisions include *strategic importance*, cost, quality, and regulation.

External Acquisition Steps

1. **Systems Planning and Selection:** Identify business needs and determine whether an external acquisition is necessary.
2. **Systems Analysis:** Define system requirements, analyze existing workflows, and establish key functionalities needed.

3. **Development of a Request For Proposal (RFP)** – document outlining requirements for vendors, may include: summary of existing systems and applications, requirements for system performance and features, reliability, backup, and service requirements, criteria that will be used to evaluate proposals, timetable and budget constraints)
4. **Proposal Evaluation:** Assess vendor responses based on predefined criteria, including technical capabilities, cost, and support services.
5. **Vendor Selection:** Choose the most suitable vendor based on evaluation outcomes and proceed with contract negotiation and implementation.

Cloud Solutions: Public cloud (third-part service providers), Private cloud (operated solely for organization, or hosted internally), Hybrid cloud (combination).

Internal Acquisition Methods

System Development Methods: Organizations can adopt various approaches to internal system development, each with its own advantages and challenges.

- **Traditional Software Development Lifecycle (SDLC):** A phased "waterfall" approach that follows sequential stages—planning, analysis, design, implementation, and maintenance. While structured, it can be costly, time-consuming, and inflexible.
 - **Agile Methodology:** Focuses on iterative development with quick releases of working software. Encourages adaptability to changing requirements and continuous collaboration between stakeholders.
 - **CI/CD (Continuous Integration and Continuous Deployment/Delivery):** Automates code integration, testing, and deployment, allowing frequent and reliable software updates while minimizing manual errors.
 - **Lean Startup:** Emphasizes experimentation over detailed planning, customer feedback over intuition, and iterative design over traditional long-term development cycles.
 - **MVP (Minimum Viable Product):** A strategy where a product is developed with just enough features to attract early adopters and validate the idea before investing in full-scale development.
 - **Prototyping:** A trial-and-error approach used to explore system functionality. It is particularly useful when requirements are unclear or when iterative feedback from stakeholders is necessary.
 - **DevOps:** A collaborative approach that integrates software development and IT operations. It follows agile principles and promotes automation, continuous monitoring, and rapid deployment to enhance system stability and performance.
- ### System Conversion Strategies
- **Conversion:** Process of changing from old system to new system. Four main strategies
 - **Parallel strategy:** Safest, but expensive
 - **Direct cutover strategy:** Risky, but fast
 - **Pilot study strategy:** Introduces the system to only a limited area of the organization first
 - **Phased approach strategy:** Introduce the system in stage, either by functions or by organization units
 - Requires end-user training. Finalization of detailed documentation showing how the system works from both technical and end-user standpoint

5. IT Project and Change Management

Project management strategies and practices at organizational and project level. Change management challenges, pitfalls, strategies and tactics.

IT Project Portfolio Analysis

- Inventories all of the organization's information systems projects and assets.
- Strategic goals & objectives, Classification of project types:
 - **Breakthroughs:** E.g., Digital twins in factory.
 - **Incremental:** Enterprise search proj (internal productivity system).
 - **Mandatory:** Cybersecurity.
- Budget allocation and commitment to specific projects: Naturally, greater the value to business, and lower the risk, the higher the funding priority. Otherwise, fund selectively or do not fund. (Project categorization affect funding decision).
- Ultimately balance risk and return from systems investments to improve return on portfolio.

Discussion: Unilever's Digital Transformation

- **Digital Twins:** A virtual representation of an object or system that spans its lifecycle, updated from real-time data, using simulation, machine learning, and reasoning to aid decision-making.
- Unilever's digital twin solution was custom-built by their engineering team in partnership with The Marsden Group (a Microsoft partner) and hosted on Microsoft's Azure platform. However, as risk is high (high value, high risk) - currently, Unilever experiment on eight digital twins first across NA, SA, Europe, Asia, streaming data from 15 of its 300 global plants.

Factors in IT Project Portfolio Analysis

- Standardization: Applications and infrastructure are well-defined and documented.
- Centralization and demand management (Well-defined scheme for screening, categorizing, and prioritizing projects).
- Metrics in project prioritization: Strategic alignment, benefits measurement, feedback mechanisms, and advanced valuations (e.g., inclusion of qualitative option value in funding decisions).

Tools

- **Competing Investments: Weighted Multicriteria Analysis:** Used to evaluate alternatives when multiple criteria exist. Assigns weights to various features of a system and calculates weighted total scores.
- **Making the Business Case:** three types of arguments commonly made in the business case for an IS:
 - **Faith:** Based on beliefs about organizational strategy, competitive advantage, industry forces, etc.
 - **Fear:** Based on the notion that failure to implement the system will result in loss to competition or business failure.
 - **Facts:** Based on data, quantitative analysis, and indisputable factors.
- **Cost-Benefit Analysis and Total Cost of Ownership (TCO)** TCO includes direct and indirect costs (capital vs. operational costs) Includes things like training, support, maintenance costs. Capital cost usually very low (maybe 20% ?).
- Benefits can be split into tangible and intangible. Tangible benefits (cost saving): Increased productivity, lower operational costs, reduced workforce, etc. Intangible benefits: Improved asset utilization, enhanced customer experience, increased job satisfaction, etc.

The Productivity Paradox

- While costs of developing an IS are easy to quantify, tangible productivity gains are often difficult to measure.
- Challenges include measurement problems (measure effectiveness and efficiency), time lags (between investment and obsv. impacts), redistribution of resources (industry level productivity is the same - zero-sum game, if rival company up, others lose market share), and mismanagement (not utilizing IS properly).

Project Management Objectives

- A project is a planned series of related activities for achieving a specific business objective. IT/IS project management deals with five major variables: **Scope, Time, Cost, Quality, Risk..** Time hardest to meet. Classic trilemma (2 out of 3) - time, cost, quality. Factors influencing project success: Project size, project structure, experience with technology.

Planning Tools

- **Gantt Chart:** Visually represents tasks, timelines, and resource requirements. Each task as a horizontal bar whose length is proportional to the time required to complete it.
- **PERT Chart:** Graphically depicts project tasks and their interrelationships.
- **Agile Approach:** Iterative development with continuous feedback and adaptation.

Change Management

- A systematic approach to managing change at both organizational and individual levels. Activities include communication, training, and process change.
- Role of a **change agent:** Redefines configurations, interactions, job activities, and power relationships. (can be internal or external agent, no best way. E.g. NUS LMS transition - internal)
- **Sources of resistance:** Organizational level: Structural inertia, threat to power relationships, resource allocation. Individual level: Habit, fear of the unknown, economic factors.
- **Models of adoption of innovation:** Demand-driven - if benefits higher than costs, adopt. Performance, effort expectancy, Social influence, Facilitating condition. Supply-driven - resistance to innovation due to knowledge barriers. Remove barriers helps to diffuse innovation. Organizational learning, skills development.

Typical Pitfalls in Change Management

- Underestimating the magnitude of change (structural and cultural).
- Poor leadership and lack of commitment to change.
- Treating change management as a "nice to have" rather than a critical success factor. (Failing to embed change management steps in the project plan.)

Change Management Strategies and Tactics

- Define clear expectations and roles; include key managers and personnel.
- Start early and sustain efforts post "go-live."
- Constant communication of project benefits.
- Integrate change activities into the project plan.
- Budget resources specifically for change activities.
- Tactics include top management support, user involvement, training, support channels, and external consultants.

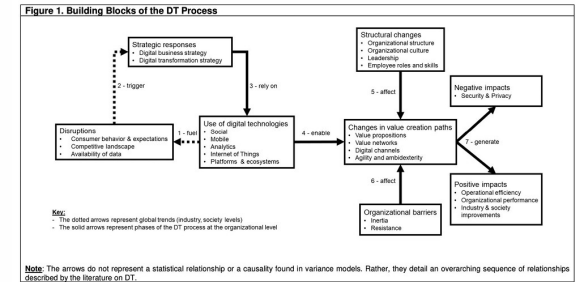
When it comes to managing and leading a team/project, eventually you are managing what? - People and personal relationships.

6. Digital Transformation and Innovation Alliance

Digital Transformation related concepts and practices, innovation related strategic alliance practices in organizations.

Digital Transformation Process

- The process of using digital technologies to create new / modify existing business processes, culture, and customer experiences to meet changing business and market requirements. It is a business transformation strategy.
- E.g. Digital twins in manufacturing, smart contracts in shipping, smart warehouses, marketing automation, Fintech digital transformation.



Source: Vial, G. (2019). 'Understanding digital transformation: A review and a research agenda'. Journal of Strategic Information Systems, 28, 118-144.

Types of Business Transformation

Business transformations can be categorized based on two questions:

1. Is your transformation driven by internal needs or external forces?
2. Does it need to happen quickly?

External force + fast/rapid adaptation pace: Hijacked transformation.

External force + slow implementation pace: Negotiated transformation.

Internal need + fast implementation pace: Sprinted transformation.

Internal need + slow implementation pace: Slow-motion transformation

E.g. DBS Bank's Digital Transformation seems to be sprinted transformation, with elements of negotiated (external).

A Typology of Innovations

Innovations classifiable into various types based on focus and impact:

- **Product innovation:** Development of a new or improved product.
- **Process innovation:** Development of a new business process.
- **Organizational innovation:** Introduce new organizational structures, systems, or practices (e.g., a new venture division, internal comm. system).
- **Management innovation:** Implementation of new management systems (e.g., Total Quality Management (TQM)).
- **Production innovation:** Adoption of new production techniques (e.g., Just-in-Time (JIT) manufacturing).
- **Commercial/Marketing innovation:** Introduction of new sales approaches or marketing strategies (e.g., omni-marketing).
- **Service innovation:** Development of new services (e.g., telemedicine).

Strategic Alliance

Contractual agreement among organizations to combine efforts and resources to meet common goal. **Reasons for forming strategic alliances:**

- Improved access to capital and new business opportunities.
- Greater technical critical mass. Shared risk, liability.
- Technology transfer benefits, Reduced R&D costs.
- Utilization of distribution skills and marketing strengths.
- Access to technology, Standardization. By-product utilization, Enhanced management skills

Types of strategic alliances: Licensing, Joint ventures, R&D consortia, Supplier innovation. Innovation networks, Industry clusters, Outsourcing, Collaboration. However, if rely too much on alliance, may de-emphasize own company's core competencies and competitive advantage.

Licensing

Licensing another firm's tech often marks beginning of collaboration.

- **Advantages:** Speed of entry into different technologies, Reduced cost of technology development.
- **Disadvantages:** Neglect of internal technology development, Risk of dependency through cross-licensing agreements.

Examples: (Cross-licensing agreements): Apple and HTC (2012): 10-year cross-licensing agreement (current & future patents). IBM and Western Digital (2016): Cross-licensing arrangement, >100 patents (distributed storage systems, nonvolatile memory devices). Google and IBM (2011): Google acquire >2,000 IBM patents (cross-licensing deals).

Licensing: Patent Pool

Patent pools collect patents from multiple patentees, license them out as a package. Common in complex, mature technologies requiring complementary and essential patents (e.g., cell networks, 5G, EV charging).

- **Benefits:** One-stop-shop licensing efficiencies, Reduced transaction costs, Facilitates innovation diffusion.
- **Anti-competitive concerns:** Pools composed solely of substitute technologies may lead to price fixing. Discrimination against non-member licensees Risk of collusion among pool participants.
- **Case Study:** Avanci (Patent Company), operator of patent licensing platforms established in 2016 in IT tech space, IoT, etc. independent intermediary at intersection of different industries. (Avanci 5G Connected Vehicle Program - span 59 initial participants. Licenses cover majority of essential cellular tech used in next-gen connected vehicles, licensors such as Qualcomm, Intel, Huawei, Samsung.)
- **Singapore CCCS Regulation:** Competition, Consumer Commission of Singapore evaluates patent pools, ensure do not distort competition. **Key considerations:** If patents in pool is essential for developing product/service. If licenses issued on non-discriminatory basis to all interested parties, if pool members remain free to license patents outside pool.
- E.g. Patent pool with all equipment and tech, establish \$300 licensing fee to be paid to pool each time group of firm's tech used → collusion.

Patent Troll

Uses patents as legal weapons, does not create new products or ideas.

- **Characteristics:** Purchases broad, non-specific patents. Engage/ threaten litigation against businesses developing innovative products. AKA. "Non-practicing entities" (NPEs).
- NPE Patent Litigation (2024): Potential benefits(?) despite negativity of setting up NPE (across companies): Pool (lowered) legal costs, shared risk in weaponizing, monetization of unused patents, avoiding counter-sue (NPE no products, purely for legal practices). Distance associated 'negative' branding from image.

Joint Venture

A separate legal entity, with partners to alliance normally being equity shareholders.

- **Benefits:** Costs and possible benefits from R&D research project shared. Potentially advantageous for companies in need of expanded resources with minimal (or no) infusion of capital. Helps with market penetration (potential collaborator in foreign market).

- **E.g.** LYNK & Co: joint venture formed in 2017 between the Chinese car brand Geely and the Swedish car maker Volvo. Ownership structure: Volvo %30, Geely Auto 50%, Geely Holdings 20% shares. Volvo Cars' minority shareholding reflects support for LYNK & CO brand. Both companies share technology, including Compact Modular Architecture (CMA) used by Volvo Cars for 40 series cars.

R&D Consortia

A consortium describes situation where multiple firms collaborate on a large-scale activity.

- **Rationale to join:** Share cost and risk of research. Pool scarce expertise and equipment. Perform pre-competitive research and setting standards.
- **E.g.** The Partnership on AI (established in 2016 by Google, Amazon, Facebook, Microsoft, etc.) to research best practices for artificial intelligence systems. BMW, Intel, and Mobileye consortium (2016) to develop an Autonomous Driving Platform.

Supplier Innovation

Accelerate innovation, companies increasingly collaborating w. suppliers.

- **E.g.:** P&G's "Connect and Develop" program: Transitioning from R&D to C&D (Connect and Develop). Schlumberger Supplier Innovation Program.
- **Challenges:** Build trust. Strategic prioritization: Identifying areas of innovation w. suppliers. Partner selection: Choose suitable innovation partners. Alignment: Defining incentives, managing risks. Infrastructure and governance: Executing and managing projects effectively.

Innovation Networks

A network that extends beyond supplier and customer relationships. Different formats. E.g. Apple Supply Chain Innovation.

Industry Clusters (by geographic proximity)

Geographic concentrations of interconnected companies, specialized suppliers, service providers, and associated institutions in a specific field.

- **Benefits of clusters:** Increase productivity of companies in the area. Drive direction and pace of innovation. Stimulate formation of new businesses within the cluster.
- Geographic, cultural and institutional proximity provides companies with special access, closer relationships, better information, powerful incentives and other advantages that are difficult to tap from a distance.

Outsourcing

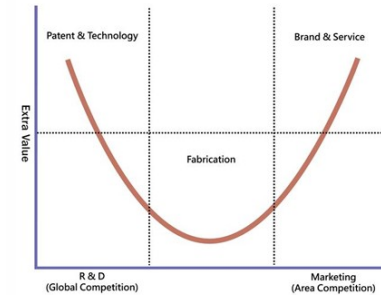
Delegation of non-core operations from internal provision to external entity specializing in that operation. Involves degree of two-way information exchange, coordination and trust between outsourcer and client.

- Evolved into a routine strategic management move affecting both peripheral functions and core competitive areas.
- **Potential Risks:** Dependence on service supplier. Hidden costs. Loss of competencies. Service provider's lack of necessary capabilities. Inefficient management. Information leakage. Social risks.

Discussion: Smile Curve

Graphical depiction of how value-added varies across different stages of bringing product to market. (in IT-related manufacturing). Suggests early and late stages of the value chain contribute more value to product/ service than middle stages. Concept proposed by Stan Shih (founder of Acer) around 1992. E.g. Acer built PCs for Dell, IBM, and Hewlett-Packard

before launching its own personal computer brand in the 1980s.



Discussion: Outsourcing Dilemma – Intel and TSMC

In July 2020, Intel considered outsourcing chip manufacturing to TSMC (Taiwan) due to production delays.

- **Key considerations:** Chip design has higher growth margin than manufacturing. Integrated manufacturing was a key competitive edge for Intel. Geopolitical risks and natural disasters (e.g., earthquakes) posed challenges. Outsourcing technical know-how overseas was considered risky.
- **Outcome:** Intel decided to partially outsource manufacturing to TSMC while keeping the majority of manufacturing in-house.
- Intel, TSMC competitors, collaborators, or something else? 'Co-Opition'.

Technology Acquisition

Strategies and practices for acquiring technology through alliances, mergers, or acquisitions. Organizational Integration of tech acquisition:



Reasons for Failure of Alliances

Common reasons include: Failure to understand, adapt to new management style required for the alliance. Lack of cultural understanding. Insufficient commitment to succeed. Divergence of strategic goals. Lack of trust. Operational or geographical overlaps. Unrealistic expectations. Case example: Hewlett-Packard (HP)'s acquisition of Autonomy.

Key Insights for Creating Value Through Strategic Alliances

Alliance Strategy, Partner Screening, Mode selection (where to ally, when to acquire). Negotiating and designing alliance (Build trust, sound governance, set milestones and exit plan). Relationship Management (Establish alliance office, keep it agile.)

7. Innovation Management

Innovation Management process and practices in organizations. Intellectual property and different types of IP.

Innovation Intensity Measures

R&D Expenditure in relation to company size, scale, others. (Not absolute).

- **Innovation spending:** Total expenditure on innovation-related activities.
- **R&D expenditures:** E.g. R&D staff compensations. Research contracts and outsourcing. Technology acquisition. Software licenses.
- **No. new products/services / patents filed / publications** in given time.
- **Measures include:** R&D expenditure-to-revenue ratio, ratio of R&D personnel to total employment etc.
- Is acquisition fee considered part of R&D expenditure? Depends on if some level R&D personnel, tech transfer, etc vs. simple business acquisition.

Innovation Types

- **Top-down innovation:** Driven by senior management. Provides focus for innovation efforts. (E.g. new ERP - enterprise resource system firm-wide.)
- **Bottom-up innovation:** Idea originate from employees. Opportunities identified through day-to-day operations. (E.g. Gmail, SoC Softmark.)

Organizational Characteristics Facilitating Innovation

1. **Growth orientation:** Long-term growth over short-term profit.
2. **Organizational heritage and innovation experience:** Widespread recognition of the value of innovation.
3. **Vigilance and external links:** Aware of threats, opportunities.
4. **Commitment to technology and R&D intensity:** Willingness to invest in long-term technology development.
5. **Acceptance of risks:** open to risky opportunities in balanced portfolio.
6. **Cross-functional cooperation and coordination:** Mutual respect among individuals and willingness to work across functions.
7. **Receptivity:** Ability to identify, well utilize externally developed tech.
8. **Space for creativity:** Manage innovation dilemma, room for creativity.
9. **Strategy towards innovation:** Strategic alignment between business objectives and innovation objectives.
10. **Coordination of a diverse range of skills:** Developing marketable products requires combining specialized knowledge.

Intellectual Property (IP)

- **IP:** creation of mind, intangible creative work, not physical form.
- Represents literary/creative works (e.g. source code), inventions - manifestations or expressions of ideas. Protected by copyright law, patent law, trademark law, and trade secret law.
- Legal protection applies to **tangible** expression of creative/original idea.

Copyright

Protects the expression of original ideas in tangible forms. Grants the copyright owner control over use and commercial exploitation.

- Reproduction, Publication, Adaptation
- Public performance, Communication to the public
- **Types of Works Covered:** Literary works (e.g., poems, source code), Dramatic works (e.g., scripts, choreography), Musical works (e.g., melodies), Artistic works (e.g., paintings, sculptures, photographs), Published editions (e.g., typographic arrangements), Sound recordings, films, broadcasts, cable programs, performances, Newer forms: software, video games, multimedia works, webpages.

Fair Use Doctrine

Enables limited use of copyrighted material without permission.

- Common uses: criticism, comment/news report, teach, research.
- Basic constitutional objective of copyright - creative progress, 'transformative' - helps to determine if is fair use of material as matter of law.
- **Four main factors considered:** *Nature* of copyrighted work (how unique/creative), *Purpose*, character of use (commercial vs. non-profit, edu), *Amount* and substantiality used, *Effect* on the market (value of work - if it harms original owner's market share).
- **Copyright Balance:** Copyright aims to protect authors while serving public interest. Courts limit copyright scope to avoid monopolistic harm to society. (E.g. - Google vs Oracle (API fair use) 11,500 lines (0.4% of 2.86M). Java SE and Android. Limit scope of copyright, ensure monopoly does not harm public interest.)

AI-Generated Works: Evolving legal status of AI-generated content.

US: gen-AI content not copyrightable, prompts insufficient for authorship.

Mostly human-authored works containing AI content still copyrightable.

SG: under development.

Patent

A patent protects inventions (new product, process, or tech improvement). Covers utilitarian or functional devices (e.g., machines, manufactured articles). **Key evaluation criteria:**

1. **Novelty:** Not known publicly anywhere. (Patent - novel, new. Vs. copyright - emphasize originality, independent work by author.)
2. **Inventive step:** Not obvious to experts in the field.
3. **Industrial application:** Must be useful and applicable in industry.

Non-Patentable Items (SG Patent Act 1994): Treatment methods for humans or animals (surgery, therapy, diagnosis). Inventions encouraging offensive, immoral, or antisocial behavior.

Other forms of IP

- **Trademark:** A sign distinguishing goods/services from others. Includes name, logo, symbol, shapes, color, sound. (E.g. Tiffany Blue®, Coke bottle shape.) **TM:** unregistered trademarks, ®: Use after official registration.
- **Trade Secret:** Confidential business information with economic value. Requires effort to develop, is unique, and not publicly known. (E.g. business operations, market research, customer data, financial affairs).
- **Singapore IP Protection Framework:** Legal Acts: Copyright Act, Patent Act, Trademark Act. Confidential information protected by common law.
- **IP Protection Duration** (SG): Copyright, ≈ 70 years, Patent 20 years, trademark 10 years renewable, trade secret as long as secrecy maintained.
- **IP Registration Requirements:** **Copyright:** No application required – automatic upon creation. **Patent:** Requires formal application. **Trademark:** Requires registration for legal protection. **Trade secret:** No registration – protection depends on confidentiality.
- **IP Ownership Considerations:** Author usually owns copyright. Exceptions: Employer owns work created during employment. Commissioned works may have specific ownership terms.

8. Risks and Security

- IT risks and security management challenges.
- Cybersecurity landscape, IT security threats, and the CIA triad.
- IT security management principles, shared security responsibility model.
- Cybersecurity legal framework, PDPA, Computer Misuse Act.

Incidents

- **2018 SingHealth Data Breach:** Cyberattack compromising personal data of 1.5 million patients. Vulnerabilities in critical information infrastructure (CII) and the importance of robust cybersecurity measures.
- **2021 Colonial Pipeline ransomware:** disrupt fuel supply (U.S. East Coast).
 - Ransomware attack that encrypted critical systems, leading to operational downtime and financial losses.
 - Paying the ransom not recommended: Incentivize further attacks, no guarantee restored access or of leaking stolen data.
- **FireEye hack:** exploited vulnerabilities in SolarWinds software to compromise multiple (US) organizations globally. Attack upstream IP supplier.

IT Security Threats

- **Supply chain attacks:** Exploiting weak links in the supply network, such as targeting software, device, or service providers. (SolarWinds Orion).
- **Data breaches:** Unauthorized access to sensitive data.
- **Account hijacking:** Compromised user credentials, unauthorized access.
- **API insecurity:** Vulnerabilities in application programming interfaces.

The C.I.A. Triad

- C.I.A. triad: 3 key characteristics of information security.
- **Confidentiality:** Access to info limited to authorized individuals.
- **Integrity:** Protect data from corrupt, damage, unauthorized modification.
- **Availability:** Ensuring data is accessible and usable when needed.
- E.g. Colonial Pipeline ransomware hack all C.I.A. compromised.

Trifecta of People, Process, and Technology

- Model emphasizes information security only as strong as weakest link among people, processes, technology. (Bruce Schneier)
- **Cannikin Law/Wooden Bucket Theory:** A bucket's capacity is determined by its shortest stave. - (People are the weakest link)
- Identifying the weakest link is crucial for strengthening overall security.

IT Security Management Principles

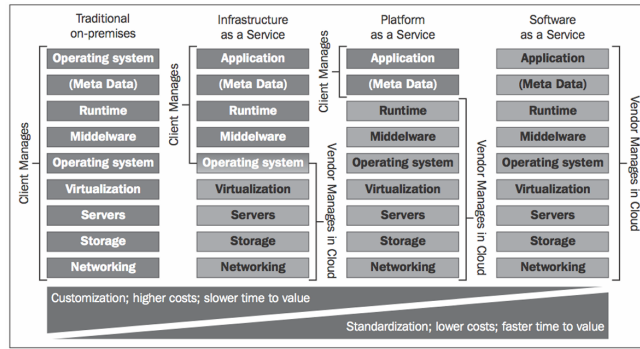
- **Need to know:** Limiting access to information based on job requirements.
- **Least privilege:** Granting users the minimum level of access (data access + modification, other rights) necessary to perform their tasks.
- **Zero-trust:** Adopt "never trust, always verify" network security approach.

Vendor Risk Management

- Ensures that third-party vendors do not introduce unacceptable risks.
- Phases of vendor risk management:
 - **Planning:** Identifying potential risks.
 - **Due diligence selection:** Evaluating vendors' security practices.
 - **Contracting:** SLAs (Service Level Agreem) define responsibilities.
 - **Monitoring and assessment:** Conducting regular evaluations, such as onsite assessments or self-evaluation reports from vendors.
 - **Termination:** Ensuring secure offboarding of vendors.

Shared Security Responsibility Model (SSRM)

- Cloud service providers adhere to shared security responsibility model. Defining these responsibilities is critical to reducing risks in public, hybrid, and multi-cloud environments
- Customer responsibilities:** Manage data, applications, configurations.
- Provider responsibilities:** Secure underlying infrastructure, e.g. physical hosts, networks.



Cybersecurity Legal Framework

Key legislations governing cybersecurity in Singapore:

- Critical information infrastructure:** (CII). Computer or comp. system located wholly or partly in Singapore, necessary for continuous delivery of essential service, loss /compromise hv debilitating effect on availability of essential service.
- (E.g. CII - Energy, Water, Banking, Healthcare, Aviation, Media, Govt, Transport - Does not mean if part of this industry it is considered. Must be of certain size / critical mass)
- Cybersecurity Act 2018:** Regulates critical information infrastructure (CII) and cybersecurity service providers.
- Personal Data Protection Act (PDPA) 2012:** Governs the collection, use, and disclosure of personal data. Key provisions include:
 - Ensure accuracy, completeness of personal data.
 - Reasonable security arrangements to prevent unauthorized access.
 - Retain personal data only as long as necessary for intended purpose.
 - Restrict transfer of personal data outside Singapore unless adequate protection ensured. (Allowed to as long as justify protected).
 - Right to Delete:** PDPA allows individuals to request deletion of personal data under certain conditions, but not absolute right. (SG - If org. can justify purpose, they may keep record forever, no right to delete).
 - Liability:** Even if company is data owner, if company has done due diligence, they may not be fined. (Data breach through third party app, zero-day vulnerability, not in their control etc.)
 - PDPA applies to only private sector organizations.** Public sector agencies (IRAS) by **Public Sector (Governance) Act (PSGA)** instead. Scope of company - e.g. SingHealth, NUS - PDPA, not PSGA.
- Computer Misuse Act (CMA) 1993:** Criminalizes unauthorized access, modification, or misuse of computer systems. Including:
 - Accessing or modifying computer material without authorization.
 - Using or intercepting computer services unlawfully.
 - Disclosing access codes or facilitating cybercrimes

9. IT/AI Governance

- Governance:** "The set of responsibilities, practices exercised by the board and executive management with goal of providing strategic direction, ensuring objectives are achieved, ascertaining that risks are managed appropriately, and verifying enterprise's resources are used responsibly."
- GRC** (Governance, Risk Management, Compliance) reflect principles.
- AI Governance:** ability to **direct, manage, monitor** AI activities of org.
- Objective** to deliver transparent and ethical AI to establish accountability, responsibility, and oversight.

AI Governance Framework

- Developed by IMDA and PDPC, Singapore's AI governance framework includes 11 AI ethical principles.
- Key Dimensions of AI Governance include:**
 - Internal Governance Structures and Measures:** Clear roles, responsibilities for ethical AI deployment (e.g., governance boards including CTO, CDO, CPO). Risk management and internal controls (e.g., MasterCard's AI governance council and risk scoring mechanisms).
 - Human Involvement in AI-Augmented Decision Making:** **Human-in-the-loop:** Humans retain full control; AI provides recommendations. **Human-out-of-the-loop:** No human oversight; AI systems have full control. **Human-over/on-the-loop:** Humans monitor AI systems and can intervene when necessary.
 - Operations Management:** Data preparation: Ensuring data quality, minimizing bias, and understanding data lineage. Good dataa accountability practices. Different datasets for train, test, validate. Periodic reviewing and updating of dataset. Algorithms: Explainability, repeatability, robustness, regular tuning, and *traceability, auditability*.
 - Stakeholder Interaction and Communication:** General disclosure: Informing stakeholders about AI use and its impacts. (E.g. know if you are chatting with AI chatbot. Option to opt-out: Allowing individuals to avoid AI-driven decisions under certain conditions (degree of harm, reversibility, availability of alternatives etc - (IMDA: Not necessary if justified/not feasible. Recommended but not mandatory)). Communication channels and acceptable user policies (AUPs).

TRANSPARENCY ON THE USE OF AI AND AI SYSTEMS Ensuring that individuals are aware and can make informed decisions			
TRANSPARENCY Appropriate info is provided to individuals impacted by AI system			
UNDERSTANDING HOW AI MODELS REACH DECISION Ensuring AI operation/results are explainable, accurate and consistent	SAFETY & RESILIENCE OF AI SYSTEM Ensuring AI system is reliable and will not cause harm	FAIRNESS / NO UNINTENDED DISCRIMINATION Ensuring that use of AI does not unintentionally discriminate	MANAGEMENT AND OVERSIGHT OF AI SYSTEM Ensuring human accountability and control
EXPLAINABILITY* Understand and interpret what the AI system is doing	SAFETY AI system safe: Conduct impact / risk assessment: Known risks have been identified/mitigated	FAIRNESS* No unintended bias: AI system makes same decision even if an attribute is changed; Data used to train model is representative	ACCOUNTABILITY Proper management oversight of AI system development
REPEATABILITY / REPRODUCIBILITY AI results are consistent: Be able to replicate an AI system's results by owner / 3rd-party.	SECURITY AI system is protected from unauthorised access, disclosure, modification, destruction, or disruption	DATA GOVERNANCE Good governance practices throughout data lifecycle	HUMAN AGENCY & OVERSIGHT AI system designed in a way that will not decrease human ability to make decisions
	ROBUSTNESS* AI system can still function despite unexpected inputs		INCLUSIVE GROWTH, SOCIETAL & ENVIRONMENTAL WELL-BEING Beneficial outcomes for people and planet

- Overall Guiding Principles:** Decisions made by AI should be explainable, transparent and fair. AI systems should be human-centric.
 - Explainable:** Explain what AI system is doing.
 - Transparent:** Right to be informed (AI is being used). Appropriate info is provided to individuals impacted by AI systems.
 - Fairness:** Ensure algorithms do not create discrimination, unjust impacts. Develop monitoring, accounting mechanisms to avoid discrimination, consult diversity of voices, demographic when develop systems.

Human Involvement in AI-Augmented Decision Making

- At this stage, many AI-augmented decision-making scenarios should still be **Human-over/on-the-loop**. CANNOT be **Human-out of-the-loop**, e.g. (AI-based disease diagnosis in healthcare, high speed financial trading, autonomous car driving).

Case Studies

- Facial recognition bias:** Asylum applicants to US, cannot utilize app to accept photos, fail to register darker skin tones. - Training Selection bias.
- Racial bias in health algorithms:** U.S. healthcare systems use of commercial algorithms, evidence of racial bias in widely used algorithms, where Black patients assigned same level of risk are sicker than White patients, reducing extra care discriminatively.
- Gender gap in hiring:** Experimental hiring engine found to penalize resumes including the word "women's" - historical bias (real data).
- HSBC:** AI Governance in All Facets of Loan Applications. Model Risk Management (MRM) framework. Follow mandatory training, keep to principles, vendor management, auditability, keep customers in loop.
- Standard Chartered:** Standard Chartered Responsible Artificial Intelligence (RAI). Inhouse Framework Responsible for Intelligence Data and Algorithm Yield (FRIDAY). Own RAI council. 5 Main engines: Explainable AI, Data Suitability, Robustness, Model monitoring and stability, Model review and validation.

Automation Paradox

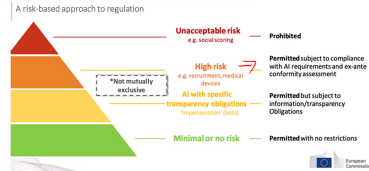
The more efficient automated systems become, the more crucial human contributions become. Humans are less involved but their involvement becomes more critical in addressing failures or unexpected events.

Other Country's AI Governance Studies

- EU AI Governance:** EU AI Act, world's first horizontal and standalone law governing AI. Three risk categories:
 - Unacceptable risk: Banned applications (e.g., social scoring, cognitive behavioral manipulation).
 - High-risk: Applications subject to specific legal requirements (e.g., biometric identification, CII, cars, medical devices, lifts, toys).
 - Minimal risk: Not explicitly banned or labelled high risk, applications left unregulated.
- US AI Governance:** National AI Initiative Act of 2020.
 - Provides a coordinated program across Federal government to accelerate AI research and application.
 - Focuses on advancing U.S. leadership in AI R&D, developing trustworthy AI systems, and preparing the workforce for AI integration
 - NIST AI Risk Management Framework (AI RMF 1.0): Principles include valid, reliable, safety, security, accountability, transparency, explainability, privacy enhancement, and fairness
- California AI Governance:** Safe and Secure Innovation for Frontier Artificial Intelligence Models Act, SB (Senate Bill) 104.
 - Intended to "mitigate the risk of catastrophic harms from AI models so advanced that they are not yet known to exist"
 - Expansive definitions of "covered models", "covered model derivatives" likely to capture many frontier AI models and subsequent modifications. SB 1047 broadly applies to "covered models," which are AI models either cost over 100 million to develop, based on covered models, fine tuned to certain extent etc.

- Apply only to companies that develop or provide compute power to train covered models or covered model derivatives, not to companies that merely use covered models.
- Before training a covered model, developers are required to implement technical and organization controls designed to prevent covered models from causing “critical harms.” (e.g. kill switch, cybersecurity etc.)
- Proposed bill may be lacking as it is classified only based on scale of economy. They should look at the nature of the AI, like the EU bill.
- **SB 53:** Focuses on whistleblower rights for employees working on foundation models.

EU AI Governance



US AI Governance



- **International Standards:** ISO/IEC JTC 1/SC 42. A joint technical committee of the International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC) focused on artificial intelligence standards.